Virtual University of Pakistan

BT403

Agriculture Biotechnology

Final term past & current papers solved

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What is synthetic seed.

• Synthetic seed can be defined as the artificial encapsulation of somatic embryo, shoot, bud, or aggregates of cells or any tissue which has the ability to form a plant in vitro.
• The use of synthetic seeds as an improvement on more traditional micropropagation protocols in facultatively propagated crops may, in the long term, have a cost saving, as the labour intensive step of transferring plants from in vitro to soil/field conditions may be overcome.

How gm can be used in medicine.

• These can be produced cheaper and easier some are: insulin, thyroid hormones and the Hepatitis B vaccine
• GM Bacteria‟s have been particularly important in producing large amounts of pure human proteins for use in medicine like clotting factors for hemophilia and human growth hormones to treat
dwarfism

What is biofertilizer Advantages of biofertilizer

• A biofertilizer (also bio-fertilizer) is a substance which contains living microorganisms which, when applied to seeds, plant surfaces, or soil, colonize the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant.
• Biofertilizers add nutrients through the natural processes of nitrogen fixation, solubilizing phosphorus, and stimulating plant growth through the synthesis of growth-promoting substances.

Advantages:

(i) They increase the yield of plants by 15-35%.
(ii) Bio-fertilizers are effective even under semi-arid conditions,
(iii) Farmers can prepare the inoculum themselves,
(iv) They improve soil texture,
(v) Bio-fertilizers do not allow pathogens to flourish,
(vi) They produce vitamins and growth promoting bio-chemical’s,
(vii) They are non-polluting

What is agrobacterium transformation

• Agrobacterium tumefaciens is a soil-borne, Gram-negative bacterium. It is rod

shaped and motile, and belongs to the bacterial family of Rhizobiaceae.
• A. tumefaciens is a phytopathogen, and is treated as the nature's most effective plant genetic engineer.

Advantages of agrobacterium mediated transformation

• As *A. tumefaciens* infects wounded or damaged plant tissues, in induces the formation of a plant tumor called crown gall.
• The entry of the bacterium into the plant tissues is facilitated by the release of certain phenolic compounds (acetosyringone, hydroxyacetosyringone) by the wounded sites.

A. tumefaciens genetically transforms plant cells and creates a biosynthetic machinery to produce nutrients for its own use.

How plant defent against pathogens.

Skoog wla b tha ek 5 marks ka Ethics long. Food med. Envi

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Write down the mineral which are added in basal media? 2 marks

What does stress with reference of plants?2

• water-logging
• drought
• lihigh or low temperatures
• excessive soil salinity
• inadequate mineral in the soil
• too much or too little light

Write down the type of biopesticides?3
1. Microbial pesticides
2. Plant-incorporated-protectants (PIPs)
3. Biochemical pesticides
4. Botanical pesticides
5. Biotic agents (parasitoids and predators)

What are the method by which plants become stress tolerant?3
• Adaptation: heritable modifications to increase fitness
• CAM plants' morphological and physiological adaptations to low H2O Environment
• Acclimation: non heritable physiological and biochemical gene expression
• Cold hardening induced by gradual exposure to chilling temps, e.g. cold-hardy Plants

How doed stock solution are prepared for tissue culture media?3

i. Macronutrient stock solution(s): Usually, the stock solution of macronutrients is prepared as 10x. Dissolve all the macronutrients one by one except (CaCl2 for macronutrient stock solution. The stock solution of CaCl2 should be prepared separately. Another way is to dissolve the different macronutrients one after the other and CaCl2 is dissolved separately and later added to the rest of the stock solution in order to avoid precipitation.

What are molecular markers?5

• Scientists can use molecular markers to select plants or animals that possess a desirable gene, even in the absence of a visible trait.
• Thus, breeding is more precise and efficient.

Types of Molecular markers

Morphological markers:
• Morphological markers are those traits that are scored visually, or morphological markers are those genetic markers whose inheritance can be followed with the naked eye.

• A molecular marker is a molecule contained within a sample taken from an organism (Biochemical molecules)
• The first biochemical molecular markers used were the protein based markers.
• One of the earliest protein based markers to be used was Isozyme.
• These are different forms of an enzyme exhibiting the same catalytic activity but differing in charge and electrophoretic mobility.

DNA based markers
• The sequence of nucleotides in DNA of an individual is unique and thus determines its identity.
• The ultimate difference between individuals lies in the nucleotide sequence of their DNA.
• These can be used to diagnose the presence of the gene without having to wait for gene effect to be seen.

What is high lysine corn 🌽?5

Under the heading of 'High-lysine com', we have described how cereals rich in certain essential amino acids such as lysine, methionine and tryptophan can be developed by genetic engineering.

• Likewise, rice is being modified into Golden rice by Prof. Inge Potrykus

and Dr. Peter Beyer.
• This is done so that vitamin A potential is maintained even after the husks are removed, a procedure adopted to allow for storage since the husks become rancid.
• This change may improve health of millions of people throughout the world.

What are ethical aspects of agricultural biotechnology with respect of food medicine and environment?10

Food
• Agriculture has always depended on plant and animal breeding and modern biotechnology provides new possibilities.
• Public perceptions of agro-food biotechnology are more critical than of its applications in healthcare. This probably results from the cultural and symbolic functions of food together with most people's relative ignorance about modern agriculture and food production.
• The most important areas where biotechnology can provide benefits for European consumers can be in improved price, quality and nutritional value of foods.

• Ethical considerations of agrofood biotechnology relate to the environment, biodiversity, sustainability, animal welfare and its socio-economic impacts.
• Consumers rights concerning biotechnological food products relate to the rights to health from safe foods, to be informed and to choose genetically engineered products or not.
• It is crucial that balanced information is provided to the public.
• Communication strategies should bring together the scientific, industrial and general communities to promote openness, dialogue and mutual understanding.

Environment

• The current environmental problems that arise from agriculture stem from modern, intensive agricultural practices and not from the use of genetically modified crops, as the latter are only currently being introduced into European agriculture.
• The use of biotechnology may either exacerbate or ameliorate these effects depending on how it is applied.
• It is therefore important, before applying new biotechnology, to consider the precautionary principle, the

need for sustainable development and the need to maintain and possibly enhance agriculturally-important biodiversity.

Medicine

• Biotechnology offers many opportunities for the production of medicines, vaccines and other medical products using agricultural sources for further improvement of human and animal health.
• New products can be developed using this technology, or the production of already existing products made more cost-effective.
• The disadvantages of the use of biotechnology must not be overlooked: although safety regulations do exist unforeseen and unwanted consequences may still occur.

What are the advantages of agrobacterium mediated transformation? 3

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Importance of hairy roots PR proteins Parasitoids Molecular Markers

Difference between modified organisms and selective breeding.

Which element is essential for growth in tissue culture.

Cell suspension culture. 5marks

 A cell suspension culture consists of cell aggregates dispersed and growing in a moving liquid media.
 It is normally initiated by transferring pieces of undifferentiated and friable callus to a liquid medium, which is continuously agitated by a suitable device.
 Suspension cultures can also be started from sterile seedlings or imbibed embryos or leaves by the mechanical method.

 Equipments:
Shaker
Centrifuge
Microscope
Haemocytometer

 Reagents and other requirements
Sterilants – alcohol, HgCl2, sodium hypochlorite All given
Nutrient medium reagents -MS basic salts and vitamins
Growth regulators - 2, 4-D
Seedlings of *Phaseolus mungo*

 Media
 MS + (1.5mg/l) 2, 4-D

Productivity losses due to stress. 5marks

Productivity losses due to stress
• Loss due to diseases range from 20 to 30 %, in case of severe infection, total crop may be lost.
• Estimated global loss due to insect pests in potential yields of all crops is ~14%.
• Losses due to insect pests ranges from 10 to 20 %
• Abiotic stresses reduce average yield of crops by upto50% .
• Most of the agricultural area is rain fed and crops in these areas invariably experience droughts of different magnitudes.
• Annually about 42% of the crop productivity is lost owing to various abiotic stress factors.

Ethical aspects of agricultural biotechnology in food, medicine and environment. 10 marks

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Artificial Seed Types of Invitro cell culture
Types of biopesticides

Principal of Anthur culture experiment
 Principle:
 Haploids refer to those plants which possess a gametophytic number of chromosomes in their sporophytes. Haploids may be grouped into two broad categories:
 (a) monoploids which possess half the number of chromosomes from a diploid species.
 (b) Polyhaploids which possess half the number of chromosomes from a polyploidy species.
 Haploid production through anther culture has been referred to as androgenesis while gynogenesis is the production of haploid plants from ovary or ovule culture where the female gamete or gametophyte is triggered to sporophytic development.

Differ b/w gm and selective breeding

Agrobacterium mediated transfusion Technological and environmental constrains of biofertilizers Ethical aspects of agriculture biotech.....

Principle of electroporation?
Ans: • Electroporation is another method of promoting competence. • In the method the cells are briefly shocked with an electric field of 10-20 kV/cm that creates holes in the cell membrane through which the plasmid DNA enters.

• This method is ready to the uptake of large plasmid DNA. • After the electric shock the holes are rapidly closed by the cell's membrane-repair mechanisms. • The efficiency with which a competent culture can take up exogenous DNA and express its genes is known as Transformation efficiency

Two source of tolerance of herbicides
Ans: • (1) Over-production of the target enzyme and
• (2) Mutations resulting in enzymes that are less sensitive to the inhibitor (usually due to a lower affinity of the enzyme for the inhibitor).

Treatments for skin pesticides?
Ans: • Rinse well with cool water. • Do not remove anything stuck to the burn. • Do not apply lotions, fats, or butter. • Do not break blisters. • Do not remove loose skin. • Cover the area with a sterile dressing, if available.

Gene splicing?
Ans: With this GM technique, biotechnologists can modify DNA, and then insert it into target host cells to allow for genes and resulting traits to be modified.
• An enzyme is then used to fuse the newly added gene into the chromosome.

Multipartite viruses?
Ans: The genome in these viruses is divided into small RNAs which may be in the same particle or different particles, e.g. brome mosaic virus (BMV). HMV contains four RNAs divided between three particles. Plant RNA viruses, in general, are characterized by high level of gene expression, good efficiency to infect cells and spread to different tissues. But the major limitation to use them as vectors is the difficulty of joining RNA molecules in vitro.

Competent cells?
Ans: Competent Cells are more likely to incorporate foreign DNA if their cell walls are altered so that DNA can pass through more easily. • Such cells are said to be competent
Types of competence
• Natural competence • Artificial competence (further details in topic 26)

Lettuce why used instead of tobacco for human growing plants
Ans: Most of the plant is leaf tissue and this tissue contains the greatest number of plastids per cell • Unlike tobacco, lettuce has no toxic alkaloids that need to be removed - low purification and downstream processing costs • Lettuce is a relevant human foodstuff that can be consumed without cooking

how Gm adds more nutrition value to plants
Ans: • The GMO method can put in added nutritional value to crops that lack necessary vitamins and minerals.

• Considering that there are places in the world relying on rice or corn as their daily staple, plant genes may be added to these crops to increase their nutritional value.
• This would help malnourished populations receive more nutrients from their diet.

tissue culture types (long)
Ans: • Types of in vitro cultures 1. Callus Cultures 2. Cell Suspension Culture 3. Anther/Microspore Culture 4. Protoplast Culture 5. Embryo Culture 6. Meristem C
Callus Culture
• An unorganized mass of thin-walled parenchyma cells
• Formation of wound callus is observed in all groups of living organisms
• In Plant tissue culture, callus is produced by cultivation of plant tissue on nutrient media under in vitro conditions
• Presence of growth hormones in the culture media promotes callus formation and proliferation
Callus: Nicotiana tabacum
Role of Callus
• In vitro callus provides totipotent cells for plant regeneration via organogenesis or somatic embryogenesis
• Callus is used as a target tissue for genetic transformation
• Callus formation is initiated for plant regeneration of other transformed tissues
• Dispersal of friable callus into single cells is used for the initiating cell suspension cultures
Initiation and maintenance of callus cultures
• (1) Selection of suitable parent material
• (2) Choice of explant and method of isolation
• (3) Culture medium and conditions required
• (4) Optimization of culture conditions
Selection of suitable parent material
• Parent plant must be healthy and free from decay or disease
• Mother plant be actively growing and not be about to enter a period of dormancy
Choice of explant and method of isolation
• Any part of mother plant; Plant organs or specific plant tissue or plant cells • Explant must contain living cells • Younger tissue is more callus responsive due to the presence of large no. of actively dividing cells • Explants isolated in sterile conditions • Ensure proper sterilization methods for a particular explant
Culture medium and conditions required
• Culture the sterilized explants on suitable autoclaved culture medium
• Incubate cultures at 22-24 ± 2°C in light or dark
• Most callus will be initiated from the cut surfaces within 3-8 wks.
Optimization of culture conditions
• Consult literature to know previous callus initiation attempts for species under consideration • For pioneering callus culture attempt, modify medium previously used for a related species • Start with one of the defined media and manipulate hormone concentrations • Set up a growth Latin Square of 25 culture plates with 5 each of auxin and cytokinin conc. • Callus for this growth trial should be uniform and of large (20 mg) size.
Callus growth measurement

• Subculture vigorous growing callus (2-5 mm dia.) • Slow growing callus plus explant transferred to fresh medium • Callus growth assessment via fresh and dry wets. • Plot a growth curve.

pesticide poisoning types

Ans: Types of pesticide poisoning:
• There are two types of pesticide poisoning:
Acute poisoning
• This happens when someone has been exposed to a high dose of pesticide.
• This could occur when the pesticide is being mixed, for example, or if a hose breaks drenching the person or bystanders with liquid pesticide solution.
• Another example might be accidental ingestion of a pesticide, such as a child swallowing the chemical.
Chronic poisoning
• This results from a person being exposed to a small amount of pesticide on many occasions over a long period of time.
• Chronic poisoning may happen when the operator repeatedly uses pesticide improperly, especially if they does not wear protective clothing and equipment or wears protective clothing which is not clean or is worn out, like wearing cracked or torn gloves.

Meristem culture principle

Ans: • Meristem culture is the in vitro culture of a generally shiny special dome-like structure measuring less than 0.1 mm in length and only one or two pairs of the youngest leaf primordia, most often excised from the shoot apex.
Principle
• The excised shoot tips and meristem can be cultured aseptically on agar solidified simple nutrient medium or on paper bridges dipping into liquid medium and under the appropriate condition will grow out directly into a small leafy shoot or multiple shoots.
• Alternatively the meristem may form a small callus at its cut case on which a large number of shoot primordia will develop.

losses of plants due to diseases

Ans: The most prominent biotic stress that the plants experience is due to disease outbreaks. • Damages lead by insect interference is also gaining a pick position regarding the biotic stress of plants. • A plant Disease may be defined as the series of invisible and visible responses of plant cells or tissues to a pathogenic microorganism or an environmental factor that results in adverse changes in form, function & integrity of plants and may lead to partial impairment or death of the plant or its parts

Giberlliene advantages

Ans: Gibberellins, although found in all plants and fungi, are unevenly distributed. The sites of synthesis are very young, unopened leaves, active buds, root tips and embryos. There are various gibberellin compounds with a similar gibbane nucleus but differ in the quality and position of the substituents of the nucleus. Gibberellic acid (GA3) is the most widely used compound. Gibberellins are normally used in plant regeneration.

GA3 is essential for meristem culture of some species. In general, gibberellins induce elongation of internodes and the growth of meristems or buds in vitro.
In its absence, the culture appears globular, due to the accumulation of nodes. Gibberellins usually inhibit adventitious root as well as shoot formation.
During organogenesis, gibberellins are antagonistic. They seem to oppose the

phenomenon of dedifferentiation.
Thus, in in vitro cultures, they cannot be used for this purpose, but can be utilized for explants already organized (meristems, apices, buds).

Name of antioxidant enzymes

- Anionic peroxidases Chilling, high CO2
- Ascorbate peroxidase Drought, high CO2, high light intensity, ozone, paraquat
- Catalase Chilling
- Glutathione Chilling drought, irradiation, heat stress, high CO2, ozone, SO2
- Glutatione reductase Chilling, drought, high CO2, ozone, paraquat
- Polyamines

Lab animals tested with gm food

Lab animals tested with gm foods

- Stunted growth
- Impaired immune systems
- Bleeding stomachs
- Impaired blood cell development
- Misshapen cell structures in the liver, pancreas, and testicles
- Altered gene expression and cell metabolism
- Their offspring have a lower

- chance of survival
- Higher blood sugar
- Enlarged livers, pancreases
- Inflamed kidneys
- Less developed brains and testicles
- Intestines reduced digestive enzymes
- The animal‟s life spans are shortened
- Inflamed lung tissue

Sequestering of herbicides resistance

- Some plants are capable of restricting the movement of foreign compounds (herbicides) within their cells or tissues to prevent the compounds from causing harmful effects.
- In this case, an herbicide may be inactivated either through binding (such as to a plant sugar molecule) or removed from metabolically active regions of the cell to inactive regions, the cell wall, for example, where it exerts no effect.

Effects of Heat Stress

• Dehydration.
• Yellowing of plants and leaves; chlorophyll deteriorates.
• Withered leaves
• Sunburn; reddish-purple tint or freckles on leaves and pseudo bulbs, leaf tip and root may turn brown.
• Alteration of gene expression is the major cause of heat stress.
• Leathery leaves; damage on cellular level.
• In response to high temperature all organisms, including plants, synthesize a set of proteins called as heat shock proteins (HSPs) The induction of HSPs at permissive temperatures have been associated with the acquisition of thermotolerance to withstand the stress

Roundup Ready crops

• 'Roundup Ready' soya:
most widely grown
GM crop variety
• tolerates applications
of Monsanto's 'broad
spectrum' glyphosate
herbicide
• Roundup destroys all
other plants.

• Italian mouse trial:
Roundup Ready soya
affects key body organs
• FSA human feeding trial:
entire transgenic gene in
GM soy survives the
passage through the
stomach and small
intestine, though not
through the colon
• Roundup residues cause
cell damage

techniques of plant tissue culture

• Culture vessels
• Culture medium
• Sterilization
• Inoculation
• Incubation
• Induction of callus

• Morphogenesis = Organogenesis Embryogenesis
• Hardening

Bio Compost:
• It is one of the Eco-friendly product composed of waste material released from sugar industries which are decomposed.
• It is magnified with human-friendly bacteria, fungi, and various plants.

Culture of protoplasts
• Protoplasts cultured in suitable nutrient media first generate a new cell wall
• The formation of a complete cell with a wall is followed by an increase in size, number of cell organelles, and induction of cell division
• The first cell division may occur within 2 to 7 days of culture
• Resulting in small clumps of cell, also known as micro colony, within 1 to 3 weeks
Importance of Protoplast Culture
• The protoplast in culture can be regenerated into a whole plant.
• Hybrids can be developed from protoplast fusion.
• It is easy to perform single cell cloning with protoplasts.
• Genetic transformations can be achieved through genetic engineering of protoplast DNA.
• Protoplasts are excellent materials for ultra-structural studies.